BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

SECTION – A (Version-C)

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q. If the charge is at rest in magnetic field, then force on charge is

(A) qvB
(B) qvBsinθ
(C) Zero
(D) qvBcosθ

Q. The principle of dynamically induced emf is used in a

(A) Choke
(B) Transformer
(C) Thermocouple
(D) Generator

Q. In a choke coil the reactance X_L and resistance R are related as:

(A) X_L = R
(B) X_L << R
(C) X_L >> R
(D) X_L = ∞

Q. An alternating voltage is given by 20 sin157t. The frequency of alternating voltage is;

(A) 25Hz
(B) 50Hz
(C) 100Hz
(D) 1000Hz

Q. A proton is about 1840 times heavier than an electron. When it is accelerated by a potential difference of 1KV, its kinetic energy will be

(A) 1840KeV
(B) 1/1840 KeV
(C) 1KeV
(D) 920KeV

Q. When a wire is stretched and its radius becomes r/2, then its resistance will be

(A) 0
(B) 2R
(C) 16R
(D) 8R

Q. Step-up transformer has a transformation ratio of 3/2. What is the voltage in secondary coil, if the voltage in primary coil is 30V?

(A) 45V
(B) 15V
(C) 90V
(D) 300V

Q. The torque in the coil can be increased by increasing

(A) Current, magnetic field, turns and Area of coil
(B) No of turns only
(C) Current and magnetic field
(D) Area of coil

Q. By doubling the absolute temperature of blackbody, the total energy radiated will change by a factor of

(A) 0.5
(B) 2
(C) 4
(D) 16

Q. In order to achieve high accuracy, the slide wire of a potentiometer should be:

(A) As short as possible
(B) As long as possible
(C) Very thin
(D) Very thick

Q. A wire is stretched to double of its length the strain is:

(A) 2
(B) 1
(C) Zero
(D) 0.5

Q. In He-Ne LASER, the discharged tube is filled with

(A) 85% He & 15% Ne
(B) 50% He & 50% Ne
(C) 15% He & 85% Ne
(D) 99% He & 1% Ne

Q. If x-rays are scattered from protons instead of electrons, the change in its wavelength for a given angle of scattering

(A) Increases
(B) Decreases
(C) Is unchanged
(D) is one radian

Q. The quantity of U^{235} in naturally occurring uranium is about

(A) 99%
(B) 63%
(C) 0.7%
(D) 27%

Q. The shear modulus of elasticity G is given by

(A) A1/Fθ
(B) F1/Aθ
(C) F/Aθ
(D) A1/F

Q. If both the length and radius of the rod are doubled, then the modulus of elasticity will

(A) Increases
(B) Decreases
(C) Doubled
(D) Remains the same

Q. When transistor are used in digital circuits, they usually operate in the

(A) Active region
(B) Saturation and cut off regions
(C) Breakdown region
(D) Linear region

Q. In a transistor both collector base and base emitter junctions are reversed biased then it is

(A) Cut off region
(B) Active region
(C) Saturation regions
(D) Q-points

SECTION – B

Maximum Marks: 40

Note: Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. What is the relationship between potential difference and electric potential energy?

ii. When Wheatstone Bridge is balanced, then no current flows through the galvanometer, why?

iii. How eddy currents can be minimized in transformer?

iv. How does a current carrying coil behave like a bar magnet? Discuss.

v. In Faraday's experiment, what would be the advantage of using coils with many turns.

vi. Sketch a graph of e.m.f. induced in an inductive coil against rate of change of current. what is the significance of the gradient?

vii. In the peak value of sine wave is 1000volts, what is the effective value?

viii. How can the conductivity of a semiconductor be raised?

ix. Why CE configuration is widely used in amplifier circuits?

x. If an electron and a proton are accelerated from rest through the same potential difference. Which particle has the longer wavelength?

xi. Why do solids give rise to continuous spectrum while hot gases give rise to line spectrum?

xii. If the potential difference in an X-ray tube is increased, how does this effect the wavelengths (a) bremsstrahlung X-rays and (b) characteristics X-ray?

xiii. Why is a conventional fission nuclear reactor not able to explode as a bomb?

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3. Q.No:3.

a) State and explain Coulomb's Law. Do include the case when the charges are placed in dielectrics. Discuss how the unit of charge "Coulomb" is defined?

(b) How a 5 mA, 100 Ω galvanometer is converted into 20V voltmeter

Q4. Q.No:4.

a) Derive the expression for the force acting on a current carrying conductor in a uniform magnetic field.

(b) A pair of adjacent coils have a mutual inductance of 1.5H. If the current in one coil changes from 0A to 20A is 0.5sec, what is the change of flux linkage with the other coil?

Q5. Q.No:5.

a) What is Laser? Explain the principle and operation of a Laser.

(b) A cylindrical steel rod 0.5m long and 1cm in radius is subjected to a tensile force of 1×10^4N. (a) What is the tensile stress? (b) What is tensile strain? (c) By what amount does the rod stretch?

Q6. Q.No:6.

Define and explain any two of the following (i) Full wave Rectification (Center-tap design) (ii) Pair production (iii) Spectrum of Hydrogen atom

GROUP – II

SECTION-A (VERSION A)

Time Allowed: 20 minutes | **Maximum Marks:** 18

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q. Some charge is being given to the conductor then its potential is:

(A) Maximum at surface
(B) Maximum at center
(C) Remains constant
(D) None of these

Q. A one microfarad capacitor of a TV is subjected to 4000 V potential difference. The energy stored in capacitor is:

(A) 8J
(B) 16J
(C) 0.004J
(D) 0.002J

Q. 10^6 electrons are moving in a conductor in one second, the current developed in the conductor is:

(A) 1.6×10^{-19}A
(B) 1A
(C) 1.6×10^{-13}A
(D) 10^6A

Q. A wire of uniform cross section A, length L and resistance R is cut into two equal halves, the resistivity of each half will be:

(A) Doubled
(B) Remains same
(C) Become half
(D) None of these

Q. The magnetic flux will be maximum for an angle of:

(A) 0°
(B) 60°
(C) 90°
(D) 180°

Q. The force on a charge particle moving parallel to magnetic field is [illegible in source scan]

(A) Maximum
(B) Minimum
(C) Zero
(D) None of these

Q. In step-up transformer _____ remains constant:

(A) Heat
(B) Current
(C) Voltage
(D) Power

Q. The emf induced in the stationary conductor placed in moving magnetic field is called _____ emf.

(A) Motional
(B) Dynamically induced
(C) Statically induced
(D) Negative

Q. The phase difference between the current and voltage at resonance is:

(A) 0
(B) π
(C) -π
(D) π/2

Q. _____ waves are not the example of electromagnetic waves.

(A) Radio
(B) Micro
(C) Ultrasonic
(D) X-rays

Q. Atoms are arranged in perfectly periodic manner in _____ solids.

(A) Amorphous
(B) Polycrystalline
(C) Crystalline
(D) All of these

Q. In a transistor, collector current is controlled by:

(A) Collector voltage
(B) Base current
(C) Collector resistance
(D) All of these

Q. Pair production cannot take place in vacuum because of law of conservation of:

(A) Charge
(B) Mass
(C) Momentum
(D) Energy

Q. Davisson and Germer experiment confirm _____ nature of electron:

(A) Particle
(B) Wave
(C) Dual
(D) All of these

Q. LASER is a device that produce _____ beam of light:

(A) Highly intense
(B) Monochromatic
(C) Unidirectional
(D) All of these

Q. In Bohr's model of hydrogen atom, the lowest orbit corresponds to _____ energy:

(A) Infinite
(B) Maximum
(C) Minimum
(D) Zero

Q. The unit of decay constant λ is:

(A) m
(B) ms
(C) s^{-1}
(D) s

Q. Gamma radiations are emitted due to _____ of the nucleus:

(A) De-excitation
(B) Excitation
(C) Spin motion
(D) None of these

SECTION-B

Maximum Marks: 40

Note: Q.No.2. Attempt any 10 parts of the following. All parts carry equal marks.

i. Voltages are always measured between two points. Why?

ii. Calculate the capacitance of the parallel plate capacitor?

iii. A heavy-duty battery of the truck maintains a current of 3A for 24 hours. How much charge flows from the battery during this time?

iv. Can neutron be accelerated in cyclotron? Give reason.

v. Explain the production of eddy currents in term of Lenz's law.

vi. Explain energy losses in case of transformer.

vii. Describe the principle of ECG.

viii. Describe some mechanical properties of solids.

ix. Explain the formation of depletion region in a PN-junction.

x. If an electron and a proton have the same De-Broglie wavelength, which particle has greater speed?

xi. Why we cannot see atom?

xii. Why $_{2}He^{4}$ have larger ionization energy than hydrogen (H)?

xiii. How many protons and neutrons are there in the nucleus of $_{86}Rn^{222}$. How many electrons are there in the neutral atom?

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03. Q.NO.03.

a) State and explain Ohm's law, also explain how resistance changes with temperature of ohmic conductors?

b) Calculate the resistance of the copper conductor having a length of 2Km and a cross-section of 22 mm^2. Assume the resistivity is 1.8×10^{-8} Ωm.

Q04. Q.NO.04.

a) What is R-L-C Series AC circuit? Explain under what condition this circuit behaves like R-C series AC circuit?

b) The peak voltage of an AC supply is 300V. Calculate the RMS voltage of the power supply.

Q05. Q.NO.05.

a) Write down the postulates of special theory of relativity; also explain how time dilatation occurs for an object moving with a relativistic speed?

b) Superman, who has exceptionally strong arms, throws a fast ball with a speed of 0.9c. If the rest mass of the ball is 0.5 Kg, what is its mass in flight?

Q06. Q.NO.06.

Write down a short note any TWO of the followings a) Galvanometer b) Magnetic Hysteresis c) Transistor